

SAT Math Mastery Program · Week 1, Session B

Lines, Slope & Linear Functions

Time	120 minutes: 90 min solving + 30 min self-check and error log
Target topics	slope from points/graphs/tables · forms of a line · intercepts · parallel & perpendicular · translations · meaning of slope/intercept · linear function notation
Teacher	Mr Malek Thiab · Dar Alfikr School
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Find slope from two points, a graph, a table, a standard-form equation, or a verbal rate.
- S2.** Write and convert between slope-intercept, point-slope, standard and intercept forms.
- S3.** Find x - and y -intercepts and use them (areas, conditions, context meaning).
- S4.** Parallel and perpendicular lines, including directly from $Ax + By = C$.
- S5.** Translate a line (up/down/left/right) and track slope and intercepts.
- S6.** Interpret slope and intercept in context; build $f(x) = mx + b$ from data and solve for inputs.
- S7.** Coordinate reasoning: collinearity, midpoint, perpendicular bisector, triangles cut by axes.
- S8.** Parameter conditions on lines (a constant fixes slope, intercept, perpendicularity or area).

2 · REQUIRED FACTS

- Slope = $\frac{\text{rise}}{\text{run}}$ = constant rate of change; the y -intercept is the value at $x = 0$; the x -intercept is where $y = 0$.
- Horizontal line $y = c$: slope 0. Vertical line $x = c$: slope undefined.
- Parallel $\iff$ equal slopes (different intercepts). Perpendicular $\iff$ slopes are negative reciprocals (product -1).
- A function is linear if equal steps in x give equal steps in y (check every gap in a table).
- Shifting a graph up k adds k to y ; slope never changes under a translation.
- Two points determine a line; on a graph read only points that sit exactly on grid intersections.

3 · REQUIRED FORMULAS

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y = mx + b, \quad y - y_1 = m(x - x_1), \quad \frac{x}{a} + \frac{y}{b} = 1$$

$$Ax + By = C : \quad m = -\frac{A}{B}, \quad x\text{-int } \frac{C}{A}, \quad y\text{-int } \frac{C}{B}, \quad \text{area with axes} = \frac{1}{2} \left| \frac{C}{A} \cdot \frac{C}{B} \right|$$

$$\parallel: A_1B_2 = A_2B_1, \quad \perp: A_1A_2 + B_1B_2 = 0, \quad \text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \quad f(a + d) - f(a) = md$$

4 · MINI THEORY SHEET

Choose the form by the data.

Given	Fastest route
two points	$m = \frac{\Delta y}{\Delta x}$, then point-slope with either point
slope and y -intercept	$y = mx + b$ directly
both intercepts $(a, 0), (0, b)$	$\frac{x}{a} + \frac{y}{b} = 1$, or $m = -\frac{b}{a}$
$Ax + By = C$	read $m = -\frac{A}{B}$; set $x = 0$ or $y = 0$ for intercepts
a table	m from any two rows; confirm the rest have the same m

Perpendicular in standard form. Swap the coefficients and negate one: $\perp$ to $Ax + By = C$ is $Bx - Ay = C'$; find C' from the given point.

Parallel in standard form. Keep A, B (or scale both), change only C .

Translation. Down k : $b \rightarrow b - k$, and the x -intercept moves by $\frac{k}{m}$. Same slope, always.

Meaning. In $y = mx + b$: m = change in y per 1 unit of x ; b = starting value. Units of m are (units of y)/(units of x).

Constant conditions. Turn the condition (slope, perpendicular, area, collinear) into one equation in the constant, then solve.

Whole-number contexts. Solve $f(x) = \text{target}$; if x must be a whole day/trip/session, round to the first value that *reaches* the target.

5 · Strategy Box

Shortcuts. (1) Build slope as a fraction; never convert to a decimal until the end. (2) On a graph, use two lattice points far apart. (3) Verify any final equation by plugging both original points. (4) Perpendicular: swap and negate. (5) Intercept triangle: area = $\frac{1}{2}|a||b|$, no integration of ideas needed.

Traps. $\frac{\Delta x}{\Delta y}$ inverted · sign lost in $y_2 - y_1$ vs. $x_2 - x_1$ order mismatch · reading the y -value of a point as the intercept · parallel vs. perpendicular confusion · “what does the *slope* mean” answered with the intercept · forgetting the answer must be a whole day/trip.

Calculator (Desmos). Plot points as (x, y) , add $y \sim mx + b$, drag sliders to match; use a table to test integer inputs; type equations in standard form directly; click an intersection to read exact coordinates.

Timing. Easy ≤ 1.5 min · Medium ≤ 3 · Hard ≤ 4.5 · Challenge ≤ 7 (total ≈ 80 min).

Pattern cues. “parallel” $\rightarrow$ same m · “perpendicular” $\rightarrow -\frac{1}{m}$ · “increases by ... for every ...” $\rightarrow m$ · “bounded by the axes” $\rightarrow$ intercepts · “for all a : $f(a + d) - f(a)$ ” $\rightarrow$ slope · “first day/trip that ...” $\rightarrow$ round up.

6 · GUIDED EXAMPLES

Example 1 — Two points $\rightarrow$ equation $\rightarrow$ intercepts

A line passes through $(-1, 7)$ and $(3, -1)$. Find its equation and both intercepts.

$$m = \frac{-1 - 7}{3 - (-1)} = \frac{-8}{4} = -2. \text{ Point-slope: } y - 7 = -2(x + 1) \Rightarrow y = -2x + 5.$$

y -intercept 5; x -intercept: $0 = -2x + 5 \Rightarrow x = \frac{5}{2}$. Check: $(3, -1)$: $-6 + 5 = -1$. ✓

Example 2 — Perpendicular through a point (standard form)

Line ℓ : $4x - 3y = 12$. Write the line through $(2, -1)$ perpendicular to ℓ .

Swap and negate: $3x + 4y = C'$. Plug in $(2, -1)$: $C' = 3(2) + 4(-1) = 2$. Answer: $3x + 4y = 2$.

Check: $A_1A_2 + B_1B_2 = 4 \cdot 3 + (-3)(4) = 0$. ✓

Example 3 — Meaning of slope and intercept

A gym plan costs 340 SAR after 2 months and 640 SAR after 5 months (linear). Write $C(m)$ and interpret its parts.

$$\text{slope} = \frac{640 - 340}{5 - 2} = 100. \quad C(m) = 100m + b; \quad 340 = 200 + b \Rightarrow b = 140. \quad \text{So } C(m) = 100m + 140.$$

100 = SAR per month; 140 = joining fee (cost at $m = 0$).

7 · PRACTICE SET — 24 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Write your work in the space below each problem. Circle your final answer. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

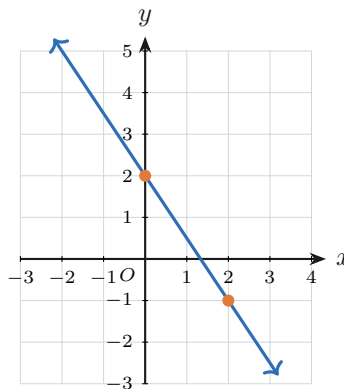
What is the slope of the line in the xy -plane that passes through $(-2, 5)$ and $(4, -7)$?

- (A) -2 (B) $-\frac{1}{2}$
(C) $\frac{1}{2}$ (D) 2

2.

EASY · DESMOS

The graph of a line is shown in the xy -plane. Which equation represents the line?



- (A) $y = \frac{3}{2}x + 2$ (B) $y = -\frac{2}{3}x + 2$
(C) $y = -\frac{3}{2}x + 2$ (D) $y = -\frac{3}{2}x - 1$

3.

EASY · NO CALCULATOR NEEDED

What is the x -coordinate of the x -intercept of the line $5x + 2y = 30$?

- (A) 6 (B) 12
(C) 15 (D) 30

4.

EASY · NO CALCULATOR NEEDED

A linear function f satisfies $f(3) = 7$, and $f(x)$ increases by 5 for each increase of 1 in x . What is $f(0)$?

5.

EASY · NO CALCULATOR NEEDED

The equation $y = 45x + 120$ gives the total cost y , in Saudi riyals (SAR), of attending x tutoring sessions. Which statement is the best interpretation of 45?

- (A) The number of sessions attended
(B) The registration fee, in SAR
(C) The cost, in SAR, of each session
(D) The total cost, in SAR, of the first session

6.

EASY · NO CALCULATOR NEEDED

Which equation represents the line that passes through $(1, 2)$ and is parallel to $y = 3x - 5$?

- (A) $y = -\frac{1}{3}x + \frac{7}{3}$
(B) $y = 3x - 1$
(C) $y = 3x - 5$
(D) $y = 3x + 2$

7.

MEDIUM · DESMOS

In the xy -plane, the line ℓ is perpendicular to the line $2x + 5y = 10$ and passes through $(4, 3)$. What is the y -coordinate of the y -intercept of ℓ ?

8.

MEDIUM · NO CALCULATOR NEEDED

The table shows three values of a linear function f . What is the value of p ?

x	2	5	9
$f(x)$	11	20	p

9.

MEDIUM · NO CALCULATOR NEEDED

A line has x -intercept -4 and y -intercept 6 . Which of the following points lies on the line?

(A) $(2, 3)$ (B) $(2, 6)$ (C) $(2, 9)$ (D) $(2, 12)$ **10.**

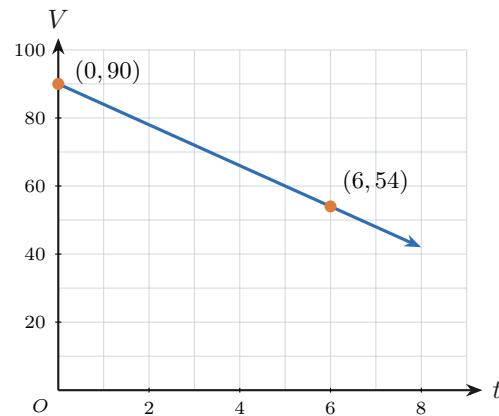
MEDIUM · DESMOS

The graph of $y = -2x + 7$ in the xy -plane is translated down 5 units. What is the x -coordinate of the x -intercept of the new graph?

11.

MEDIUM · CALCULATOR

A tank at a desalination plant is being drained. The graph shows the volume V of water, in thousands of liters, t minutes after draining begins. If the volume keeps decreasing at the same rate, after how many minutes will the tank be empty?



12.

MEDIUM · NO CALCULATOR NEEDED

For the line $kx + 3y = 12$ in the xy -plane, k is a constant. If the slope of the line is -2 , what is the value of k ?

- (A) -6 (B) -2
(C) 2 (D) 6

13.

MEDIUM · NO CALCULATOR NEEDED

A line in the xy -plane passes through $(-1, 4)$ and $(3, 12)$. What is the y -coordinate of the point on the line with x -coordinate 10?

14.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the line $2x + 3y = 12$ and the coordinate axes bound a triangle. What is the area of the triangle?

- (A) 6 (B) 12
(C) 24 (D) 36

15.

MEDIUM · NO CALCULATOR NEEDED

The points $(1, 2)$, $(4, 8)$, and $(k, 20)$ lie on the same line in the xy -plane. What is the value of k ?

16.

MEDIUM · NO CALCULATOR NEEDED

The functions f and g are defined by $f(x) = 5x + 12$ and $g(x) = 2x - 9$. The function h is defined by $h(x) = f(x) - g(x)$. What is the x -coordinate of the x -intercept of the graph of $y = h(x)$?

17.

HARD · DESMOS

In the xy -plane, points $P(-2, 1)$ and $Q(6, 5)$ are the endpoints of a segment. Which equation represents the perpendicular bisector of $\overline{PQ}$?

(A) $y = \frac{1}{2}x + 2$

(B) $y = -\frac{1}{2}x + 4$

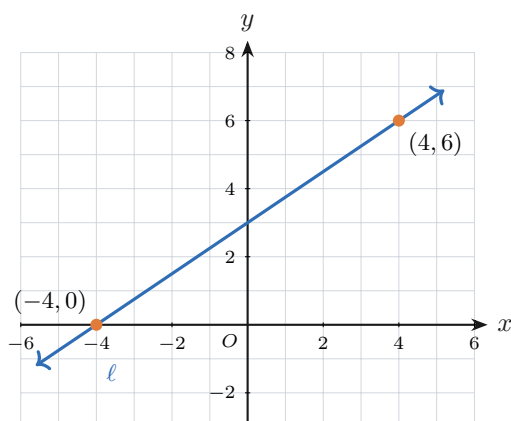
(C) $y = 2x - 1$

(D) $y = -2x + 7$

18.

HARD · DESMOS

In the xy -plane, line ℓ is shown. Line m is perpendicular to ℓ and passes through the y -intercept of ℓ . What is the area of the triangle bounded by ℓ , m , and the x -axis?



19.

HARD · DESMOS

A line in the xy -plane passes through $(2, 3)$. Its x -intercept is twice its y -intercept, and neither intercept is 0. What is the x -intercept of the line?

20.

HARD · NO CALCULATOR NEEDED

For a linear function f , $f(a + 2) - f(a) = 6$ for every real number a , and $f(0) = -1$. What is $f(5)$?

(A) 11

(B) 14

(C) 29

(D) 30

21.

HARD · CALCULATOR

Nora's savings grow linearly. On day 3 she has 250 SAR and on day 9 she has 490 SAR. On what is the first day on which she has at least 1000 SAR?

22.

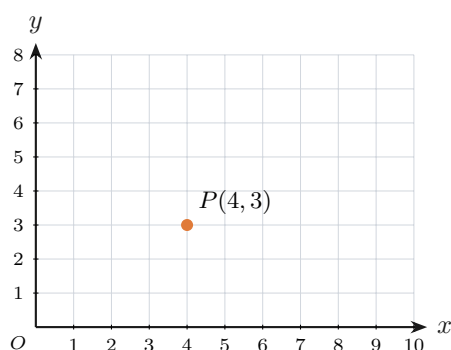
HARD · DESMOS

In the xy -plane, the lines $(k + 1)x + 6y = 5$ and $2x + ky = 9$ are perpendicular. What is the value of k ?

23.

SAT CHALLENGE · DESMOS

In the xy -plane, a line with negative slope passes through the point $P(4, 3)$ shown. Together with the positive x - and y -axes it forms a right triangle of area 24. What is the slope of the line?



24.

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear function $f(x) = ax + b$ satisfies $f(f(x)) = 9x + 8$ for all real x . What is the greatest possible value of $f(1)$?

8 · HINT SECTION — HINTS ONLY, NO SOLUTIONS

1. Subtract in the same order on top and bottom.
2. Read the y -intercept, then count rise over run to the next lattice point.
3. At an x -intercept, $y = 0$.
4. Move from $x = 3$ to $x = 0$: three steps of -5 .
5. Slope = change in cost per one more session.
6. Same slope; use point-slope with $(1, 2)$.
7. Slope of the given line is $-\frac{2}{5}$; take the negative reciprocal.
8. Get the slope from the two complete columns.
9. Slope = $-\frac{b}{a}$ with $a = -4$, $b = 6$; then $y = mx + 6$.
10. New equation: subtract 5 from the right side.
11. Rate = $\frac{\Delta V}{\Delta t}$ between the two labeled points; then solve $V = 0$.
12. Solve for y , or use $m = -\frac{A}{B}$.
13. Slope first; then move 11 units in x from $x = -1$.
14. Find both intercepts; they are the legs.
15. Equal slopes between any two pairs.
16. Simplify $h(x)$ first; its x -intercept is where $h(x) = 0$.
17. Midpoint first; then the negative reciprocal of the segment's slope.
18. Read the two labeled points; get ℓ , then m ; the x -axis chord is the base.
19. If the y -intercept is b , the x -intercept is $2b$; use $\frac{x}{2b} + \frac{y}{b} = 1$.
20. The difference over an x -step of 2 is $2m$.
21. Model $f(d)$, solve $f(d) \geq 1000$, then choose the first whole day.
22. For $A_1x + B_1y = C_1$ and $A_2x + B_2y = C_2$, perpendicular means $A_1A_2 + B_1B_2 = 0$.
23. With intercepts a, b : $ab = 48$ and $\frac{4}{a} + \frac{3}{b} = 1$. Eliminate one unknown.
24. Expand $f(f(x))$ and match coefficients: two equations, two cases for a .

9 · FULL SOLUTIONS

1. **Answer: (A).** $m = \frac{-7-5}{4-(-2)} = \frac{-12}{6} = -2$.
2. **Answer: (C).** y -intercept 2. From $(0, 2)$ to $(2, -1)$: rise -3 , run 2, so $m = -\frac{3}{2}$. $y = -\frac{3}{2}x + 2$. (D) reuses -1 as the intercept.
3. **Answer: (A).** $5x = 30 \Rightarrow x = 6$.
4. **Answer: -8 .** $f(0) = 7 - 3(5) = -8$.
5. **Answer: (C).** 45 is the slope: each additional session adds 45 SAR. (120 is the registration fee.)
6. **Answer: (B).** $m = 3$: $y - 2 = 3(x - 1) \Rightarrow y = 3x - 1$. (A) is perpendicular; (C) is the given line itself; (D) reuses 2 as the intercept.
7. **Answer: -7 .** $m_{\perp} = \frac{5}{2}$. $y - 3 = \frac{5}{2}(x - 4) \Rightarrow y = \frac{5}{2}x - 7$. y -intercept: -7 .
8. **Answer: 32.** $m = \frac{20-11}{5-2} = 3$. $p = 20 + 3(9 - 5) = 32$.
9. **Answer: (C).** $m = \frac{6-0}{0-(-4)} = \frac{3}{2}$, $y = \frac{3}{2}x + 6$. At $x = 2$: $y = 9$. So $(2, 9)$.
10. **Answer: 1.** $y = -2x + 2$; set $y = 0$: $x = 1$.
11. **Answer: 15.** $m = \frac{54-90}{6-0} = -6$. $V = 90 - 6t = 0 \Rightarrow t = 15$ minutes.
12. **Answer: (D).** $3y = -kx + 12 \Rightarrow m = -\frac{k}{3} = -2 \Rightarrow k = 6$.
13. **Answer: 26.** $m = \frac{12-4}{3+1} = 2$. $y = 4 + 2(10 - (-1)) = 26$.
14. **Answer: (B).** Intercepts 6 and 4. Area $= \frac{1}{2}(6)(4) = 12$.
15. **Answer: 10.** $m = \frac{8-2}{4-1} = 2$; $\frac{20-2}{k-1} = 2 \Rightarrow k - 1 = 9 \Rightarrow k = 10$.
16. **Answer: -7 .** $h(x) = 3x + 21 = 0 \Rightarrow x = -7$.
17. **Answer: (D).** Midpoint $(2, 3)$; slope of $PQ = \frac{4}{8} = \frac{1}{2}$; perpendicular slope -2 . $y = -2x + 7$. (A) is line PQ ; (B) drops the reciprocal; (C) drops the negative.
18. **Answer: $\frac{75}{8}$.** ℓ : $m = \frac{6-0}{4+4} = \frac{3}{4}$, $y = \frac{3}{4}x + 3$. $m \perp \ell$ through $(0, 3)$: $y = -\frac{4}{3}x + 3$, x -intercept $\frac{9}{4}$. Vertices $(-4, 0)$, $(\frac{9}{4}, 0)$, $(0, 3)$: base $\frac{25}{4}$, height 3; area $= \frac{1}{2} \cdot \frac{25}{4} \cdot 3 = \frac{75}{8} = 9.375$.
19. **Answer: 8.** $\frac{2}{2b} + \frac{3}{b} = 1 \Rightarrow \frac{4}{b} = 1 \Rightarrow b = 4$. x -intercept = 8. ($y = -\frac{1}{2}x + 4$ passes through $(2, 3)$.)
20. **Answer: (B).** $2m = 6 \Rightarrow m = 3$, $f(x) = 3x - 1$, $f(5) = 14$. (Not 29: that uses $m = 6$.)
21. **Answer: 22.** $m = \frac{490-250}{9-3} = 40$; $f(d) = 40d + 130$. $40d + 130 \geq 1000 \Rightarrow d \geq 21.75$. First whole day: 22.
22. **Answer: $-\frac{1}{4}$.** $2(k + 1) + 6k = 0 \Rightarrow 8k = -2 \Rightarrow k = -\frac{1}{4}$. (Check by slopes: $-\frac{k+1}{6} \cdot (-\frac{2}{k}) = \frac{k+1}{3k} = -1$.)
23. **Answer: $-\frac{3}{4}$.** $\frac{4}{a} + \frac{3}{b} = 1 \Rightarrow 4b + 3a = ab = 48$. With $b = \frac{48}{a}$: $\frac{192}{a} + 3a = 48 \Rightarrow 3a^2 - 48a + 192 = 0 \Rightarrow (a - 8)^2 = 0$, so $a = 8$, $b = 6$. Slope $= -\frac{b}{a} = -\frac{3}{4}$. (The double root shows 24 is the *smallest* possible area for a line through P .)
24. **Answer: 5.** $f(f(x)) = a^2x + ab + b$. So $a^2 = 9$ and $b(a + 1) = 8$. $a = 3 \Rightarrow b = 2$, $f(1) = 5$. $a = -3 \Rightarrow b = -4$, $f(1) = -7$. Greatest: 5.